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A Typologically Motivated Analysis of Projective Contents in Bura-Pabir

ORTHOGRAPHIC GUIDELINES

The following orthographic conventions were utilized¹:

<a> /a/
 /b/
<ɓ> /ɓ/
<ch> /tʃ/
<d> /d/
<dz> /dz/
<dʒ> /dʒ/
<e> /e/
<f> /f/
<g> /g/
<gh> /ɣ/
<h> /h/
<i> /i/
<ə> /i/
<j> /dʒ/
<k> /k/
<kh> /x/
<k'> /k'/*
<l> /l/
<m> /m/
<n> /n/
<ng> /ŋ/
<ny> /ɲ/
<o> /o/
<p> /p/
<r> /r/
<s> /s/
<sh> /ʃ/
<t> /t/
<ts> /ts/
<u> /u/
<v> /v/
<w> /w/
<y> /y/
<z> /z/
<zh> /ʒ/
<'> /ʔ/

*Hausa k with hook (U+0199) is often used by native speakers.

¹ Though it is beyond the scope of this paper, the issue of tone in Bura is a complicated one. Tones reported here are those heard by the transcriber; where tones are not reported, the transcriber either failed to identify them or did not feel comfortable ascribing a specific tonal measure. Sounds marked like *á* have a high tone, while *à* indicates a low tone.

1. Introduction

Bura-Pabir (bwr) is a Chadic language spoken by around 551,000 speakers, primarily those identifying as Bura-Pabir, in northeastern Nigeria^{2,3} (Eberhard et al., 2025). Bura-Pabir (henceforth Bura) is largely underdocumented and understudied, lacking an authoritative reference grammar and with only a handful of (primarily morphosyntactic) investigations associated with it. To date, there have been few serious explorations of how semantic or pragmatic meaning is generated in Bura (but see Hartmann & Zimmermann, 2012; Zimmermann, 2007). Thus, this investigation heeds the call for more cross-linguistic work on projection and projective contents articulated in Tonhauser et al. (2013) and takes Bura as a unique opportunity for a cross-linguistic comparative typology of projection. Specifically, this investigation seeks to understand how expressions in Bura give rise to projective contents along the axes presented in Tonhauser et al. (2013): that is, whether a trigger *t* of a content *m* in Bura expressions imposes a **STRONG CONTEXTUAL FELICITY CONSTRAINT (SCFC)**, whether a content *m* of trigger *t* is projective, and whether a content *m* must have its effect locally or if *m* can be satisfied in the global context (i.e., does *m* have an **OBLIGATORY LOCAL EFFECT**, or **OLE**). Given length limitations, this investigation will largely present a descriptive analysis of four select triggers and their respective contents, offering a broad overview of their behavior across different contexts and under the diagnostics provided by Tonhauser et al. (2013). To this end, I conducted 4 half-hour interviews with Yakubu, a native speaker of the Hyil Hawul dialect of Bura. These interviews were structured as formal elicitations of acceptability and indirect implication

² The author would like to thank Yakubu Bitrus for offering up his language for study.

³ The following abbreviations will be used throughout: ACC: Accusative; comp: Complementizer; compl: Completive; DAT: Dative; def: Definite; FUT: Future; gen: genitive; NEG: Negation; NOM: Nominative; pl: Plural; POSS: Possessive; prog: Progressive; SG: Singular; q: interrogative; 1: First-person; 3: Third-person

judgments; interviews were largely conducted in English. The decision to use English as a metalanguage was motivated by a lack of full proficiency in Bura on the part of the investigator, though it should be noted that the use of a non-target language as a metalanguage is not necessarily detrimental to judgement data (Matthewson, 2004). Data from Bura was also available from work done in the Fall of 2025, though these data did not consist of acceptability judgments and were instead primarily used to generate example sentences and create culturally-specific contexts. This investigation opens with an overview of the diagnostics for projection as they are articulated in Tonhauser et al. (2013) before transitioning into a description of the triggers of interest in Bura. Acceptability and indirect implication judgments are then presented for all triggers across the axes of projective content analysis (see *supra*). However, due to spatial limitations, some sentences and corresponding judgments are described in prose rather than given with a numbered example. The decision to describe in prose is wholly arbitrary, and trilinear glosses of all examples are available by request. This investigation closes by providing a summary of the behavior of projective expressions in Bura, striking a three-way comparison between English (eng), Paraguayan Guaraní (gug), and Bura, and outlining future research pathways.

2. Background: Tonhauser et al. (2013)

In their influential investigation of projective contents in Paraguayan Guaraní (henceforth Guaraní), Tonhauser et al. (2013) argue that projection judgments must be presented such that the utterance context is controlled to respect whether the expression *t* imposes a so-called STRONG CONTEXTUAL FELICITY CONSTRAINT, or **SCFC**. Specifically, projection judgments must control for whether the projective content *m* of some natural expression *t* must be entailed or implied by the utterance context for an utterance to be licit. Utterance contexts in which *m* is entailed or

implied are considered *m*-positive, whereas those which entail neither *m* nor its negation are considered *m*-neutral. If judgments are elicited absent controlling for these contextual factors, there is no way of discerning whether an utterance is unacceptable or fails to project because it isn't projective or because the context is infelicitous. To this end, Tonhauser et al. (2013) propose collecting projection judgments by first testing for whether an utterance, containing the content *m* and trigger *t*, is acceptable in contexts that either require *m* or are agnostic about *m*. If an utterance is only acceptable in the former (i.e., in the *m*-positive context), then the trigger *t* imposes a **SCFC**. These triggers are also those which deny the "availability of accommodation"; that is, a listener cannot update or alter their understanding of the context if the trigger is not licensed within an appropriate context (Tonhauser et al., 2013, p. 80). For instance, Tonhauser et al. (2013) argue that additive particle *avei* in Guaraní imposes a **SCFC** for the implication that a similar event is ongoing and thus hearing sentence such as (1) out-of-the-blue or given the context in (1) would be unacceptable such that a listener could not automatically update the common ground to accommodate the bus driver's eating of an empanada. (1) would only be acceptable if uttered in the context described in (1a) were established.

1. [Context: Malena is eating her lunch, a hamburger, on the bus going into town. A woman who she doesn't know sits down next to her and says:]

Ñande-chofeur o-karu empanáda **avei**.

"Our bus driver is eating empanadas, too."

- a. [Context: Same as (1a) except Malena is eating an empanada.]
(Example adapted by Jude Singleton from Tonhauser et al., 2013, pp. 78-79)

Given that certain utterances containing *t* with content *m* must be presented within facilitatory contexts, Tonhauser et al. (2013) argue that projection judgments, such as those elicited in response to FAMILY-OF-SENTENCES (**FoS**) variants (a common test for projection), must take care to integrate *m*-positive and *m*-neutral contexts. Likewise, Tonhauser et al. (2013)

provide diagnostics for diagnosing OBLIGATORY LOCAL EFFECT which specify whether a trigger *t* imposes a SCFC. Spatial limitations curtail the ability to reiterate these diagnostics for FoS and local effect, but Bura examples make reference to the numbering scheme used in Tonhauser et al. (2013) to direct readers to the relevant diagnostic; this is indicated in bolded brackets with the relevant number after every example.

3. Selected Triggers

The triggers and contents explored in this investigation were selected arbitrarily to represent each of the four classes of projective content proposed for English and Guaraní (see **Image 1**).

CLASSES	PROJECTION	PROPERTIES OF CONTENTS	
		STRONG CONTEXTUAL FELICITY	OBLIGATORY LOCAL EFFECT
A	yes	yes	yes
B	yes	no	no
C	yes	no	yes
D	yes	yes	no

Image 1. Proposed classificatory system of projective contents (Tonhauser et al., 2013, p. 67)

Specifically, this investigation examines the projective and local behavior, with respect to the SCFC, of the following triggers and their corresponding content(s), with the latter expressed in parentheses: the additive adverbial particle *tsúwà*, meaning ‘also, too, again’ (Class A: the *existence implication* that “there is an alternative true proposition” [Tonhauser et al., 2013, p. 72]); the expressive modifier *bzàbzà*, meaning ‘damn, spoiled, naughty, nasty’ (Class B: the *descriptive implication* that some quality is true of the referent); the demonstrative determiner *ghèndá*, meaning ‘that’ (Class D: the *indicational implication* that there is a suitable referent available & Class B: the *descriptive implication*); and the factive verb *sàndà*, meaning ‘to know’

(Class C: the implication of the content of the factive complement, i.e., the embedded presupposition). These triggers not only span the different groupings proposed in Tonhauser et al. (2013) but also capture a range of implicatures (e.g., presuppositions with the factive *sándà* and demonstrative determiner *ghèndá*, conventional implicatures with the expressive modifier *bzábzà*) and grammatical functions/lexical categories (e.g., adverbs, verbs, adjectives, and determiners).

As a prelude to the data presented herein, some basic facts about Bura morphosyntax are necessary. Bura is a SVO language and is largely analytic, tending to express tense, aspect, and mood (among other grammatical functions) as discrete words rather than marking or inflecting on the verb itself. The expressive modifier and demonstrative determiner described above are also always marked post-nominally, as are all other determiners and adjectives in Bura. Pertinent to the construction of **FoS** variants, Bura displays relatively transparent formational processes for negated and interrogative variants which do not interfere appreciably with the ability to use one trigger over another. Finally, the use of the verb *kələŋkr*, ‘believe, accept,’ in OLE judgments was chosen for its harmony with the guidelines for licit belief-predicate use in Tonhauser et al. (2013). Bura has a number of verbs describing belief across several different contexts, but Yakubu stated that *kələŋkr* is ideal for expressing belief in facts or states of affairs, such as those presented in examples (9) through (12). All sentences were double-checked for grammaticality with Yakubu during interview sessions and were cross-referenced with available literature on Bura (e.g., Blench, 2010; Hartmann et al., 2008; Hartmann & Zimmermann, 2012; Mshelbwala, 2022) and previously collected FLEx data.

4. SCFC in Bura

SCFC judgments were collected across two interviews with Yakubu. Prior to meeting, example sentences containing the relevant trigger were drafted with corresponding *m*-neutral and *m*-positive contexts; in both interview, sentences were initially presented given the *m*-neutral context and subsequently presented given the *m*-positive context if Yakubu found the first context + sentence pair unacceptable. In the case of the additive particle *tsúwà*, *tsúwà* appears to impose a SCFC, but curiously for an *m* which is not attested in either Guaraní or, from personal inferences, in English. Specifically, Yakubu stated that (2) was unacceptable given that Kucheli, who is buying milk, does not know the person buying the rice. However, the same utterance under the context described in (2a), in which the strangers from (2) are known to Kucheli, was acceptable.

2. [Context: Kucheli is in line at the store with a bottle of milk she intends to purchase. Two strangers, a man and a woman, are in the store and are standing in line next to Kucheli. The woman points to the man and says to Kucheli the following:]

Tá kwá màs shìngkáfá tsúwà

tá kwá màs shìngkáfá tsúwà

3SG.NOM prog buy rice again, also, too

“He is eating rice, also.” [#12(i)]

- a. [Context: same as (2), but the strangers are friends of Kucheli’s] **[#12(ii)]**

Tonhauser et al. (2013) report that the Guaraní equivalent to the additive particle, *avei*, is also sensitive to the utterance context, but only insofar as the context establishes the ongoing event implied by the trigger (i.e., the existence implication; see ex. (1)). For Bura, Yakubu explicitly stated that this sentence is acceptable only if Kucheli knows the other shopper and is always unacceptable if the shopper is a stranger to her. Interestingly, the use of *tsúwà* was deemed acceptable even though Kucheli and the shopper in (2a) are purchasing different items (and therefore, performing different ongoing actions); his judgement was maintained when

Yakubu was reminded of this fact. There are a handful of potential explanations for this divergence from Tonhauser et al. (2013): firstly, it may be the case that Yakubu is interpreting the fellow shopper in (2a) to be also purchasing rice. However, this still does not explain why the utterance is unacceptable in (2), where this could still be the case, but where the relationship between Kucheli and the shopper is unclear. Alternatively, it is possible that *tsúwà* conveys a salience implication (as discussed in Tonhauser et al., 2013) and that there is a requirement for the alternative action to be established in the common ground. In this sense, (2a) is acceptable because Kucheli may be perceived as more likely to have some general concept of what those known to her are doing at any given time (i.e., the common ground contains online statistics about other actants within a given context, and those actants which are known to Kucheli are more likely to be recorded as such in the common ground). To test this possibility, future elicitation sessions should probe for both a different additive particle and try to elicit judgments for utterances in contexts without human subjects, if only to avoid this nebulous implicational problem pertaining to human subjects. Tentatively, however, this investigation will argue that the additive particle *tsúwà* imposes a **SCFC**, in line with work on Guaraní for its translational equivalent of the additive particle.

The expressive modifier *bzábzá*, by contrast and in accordance with findings from Guaraní and English, does not impose a **SCFC** for the implication that the referent is as described by the modifier. That is, the use *bzábzá* is acceptable in a context such as in (3), which is neutral with respect to the attitudes of the addressee and speaker towards Musa prior to the use of the modifier.

3. [Context: Kucheli and Arhel are local students. Their teacher, Mr. Musa, is a strict but fair teacher. Musa takes Arhel into his office for a talk. Later, Arhel tells Kucheli:]

Musa bzébzá wètè ká yà kamata ki rùbùtúmta
 Musa bzébzá wètè ká yà kamata k -i rùbùtú -mta
 name expressive seen comp should comp 1SG.NOM write compl
 kákádũrna dǽpá [#12(i)]
 kákádũ -r -na dǽpá
 book gen 1SG.POSS tomorrow

“Damn nasty Musa wants me to finish writing my book tomorrow.”

In a similar vein, the factive verb *séndà* also does not impose a SCFC on the content of its complement clause. The context in (4) is neutral with respect to the addressee knowing that Arhel’s son has to organize the dishes (i.e., the presupposition embedded under the factive verb). Thus, the implication is not sensitive to the effects of context, as is true for Guaraní and English.

4. [Context: Kucheli is at home when she hears shuffling outside. She goes outside to find her neighbor’s son hiding a dirty pot in her yard. Later, Arhel apologizes and says:]

Tá séndà àbér ya kamata ka tá namǽla
 tá séndà àbér ya kamata ka tá nam -ǽla
 3SG.NOM know comp might comp 3SG.NOM make, cause thoroughly
 tùmýəri [#12(i)]
 tùm -yəri
 pot pl

“He knows that he must thoroughly organize the pots.”

Lastly, the demonstrative *ghèndá*, when used in a NP, tracks with the pattern observed in Guaraní and in English. (5) is neutral with respect to whether there is a suitable referent available for the addressee, since Arhel does not indicate one and does not share knowledge about rice with the addressee. Yakubu judged this unacceptable so long as there is no clear referent. A revised *m*-positive context in (5a) was judged as acceptable, by contrast, since this context establishes a clear referent for the demonstrative determiner. That is, the indicational implication

of *ghèndá* (that there is a suitable referent) imposes a SCFC. (5a) also indicates that the descriptive implication of *ghèndá* (that the referent bears some quality or fits some description, in this case that it is a dog) does not impose a SCFC, since Arhel does not know prior to the utterance what the referent is, though he is aware of there being a referent of some kind.

5. [Context: You are reading a book when Arhel walks over to you and says:]

Shìngkáfá ghèndá ní àrná [#12(i)]

shìngkáfá ghèndá -ní àrná

rice that def 1SG.POSS

“That rice is mine.”

- a. [Context: You and Arhel are walking through a field. Both of you can see something ahead within the grass, but it’s unclear what exactly it is. Neither of you is sure whether it is an animal, person, or object. Your vision is better than Arhel’s, and as you approach, you recognize what is lying ahead. You say to Arhel:]

Kila ghèndá ní àrná [#12(i), #12(ii)]

kila ghèndá -ní àrná

dog that def 1SG.POSS

“That dog is mine.”

Overall, the patterns observed in Bura with respect to the SCFC impositions of these four triggers largely align with judgments from both Guaraní and English. Specifically, it appears that expressive modifiers, factive verbs, and demonstrative determiners contained within NPs overlap in their sensitivity to utterance context. The exception to this overlap, the additive *tsúwà*, does not appear to impose a SCFC for the existence implication at all (given the felicity of ex. (2a)) and instead seems to be sensitive to whether an ongoing action is salient in the common ground. However, the extent to which this divergence is a consequence of salience or a spurious effect of a grammatical or lexical property of the additive remains unclear.

6. Projection in Bura

Following the guidelines set by Tonhauser et al. (2013), projective contents of the triggers under study can be assessed once their contextual sensitivity has been determined. Those triggers *t* which impose a **SCFC** with respect to their content *m* have projective implications if and only if **FoS** utterances containing *t* under an *m*-positive context are deemed acceptable (i.e., the **SCFC** must persist across **FoS** variants). For triggers which do not impose a **SCFC** or do so only for certain implications *n* (as in the case of *ghèndá* in Bura [see ex. (5) & (5a)] and the demonstrative *ko* in Guaraní), contents are projective if they are acceptable in *m*-neutral or *m*-neutral/*n*-positive contexts. The reader is referred to Tonhauser et al. (2013, p. 83) for a full set of diagnostics.

Tsúwà, in contrast to the translational equivalents in Guaraní and English, appears to project regardless of whether the context is *m*-neutral or positive with respect to the existence implication. The context in (6a-e) is neutral insofar as it does not establish or negate an ongoing event of rice-buying; **FoS** variants were generally judged acceptable, though with some exceptions. These inconsistencies, perhaps suggestive of *tsúwà* imposing a **SCFC**, disappeared in an *m*-positive context (7a-e), where Kucheli is buying rice. Importantly, this *m*-positive context *did not* control for whether Kucheli knew the stranger who was buying the rice – given this, the driving factor behind the diverging judgments of examples (2), (2a), (6a-e), and (7a-e) may reside in the grammatical function of *tsúwà* itself. More controlled contexts and extended judgement elicitation sessions, preferably with other speakers, are needed to clarify this picture. Ultimately, this investigation argues that *tsúwà* has unclear projective behavior and dog-ears the trigger for further research.

6. [Context: Kucheli is in line at the store with a bottle of milk she intends to purchase. Two strangers, a man and a woman, are in the store and are standing in line next to Kucheli. The woman points to the man and says to Kucheli the following:]

a. #⁴ Tá kwá màs shìngkáfá tsúwà [#37(i)]

tá kwá màs shìngkáfá tsúwà

3SG.NOM prog buy rice again, also, too

“He is buying rice, also.”

b. Tá adí kwá màs shìngkáfá tsúwà wá [#37(i)]

tá adí kwá màs shìngkáfá tsúwà wá

3SG.NOM not prog buy rice again, also, too NEG

“He is not buying rice, also.”

c. Tá kwá màs shìngkáfá tsú yá ? [#37(i)]

tá kwá màs shìngkáfá tsú yá

3SG.NOM prog buy rice again, also, too q

“Is he buying rice, also?”

d. Tá tsokta màs shìngkáfá tsúwà [#37(i)]

tá tsokta màs shìngkáfá tsúwà

3SG.NOM modal buy rice again, also, too

“He could be buying rice, also.”⁵

e. # Mè tá màs shìngkáfá tsúwà , yà dzi

mè tá màs shìngkáfá tsúwà yà dzi

if 3SG.NOM buy rice again, also, too 1SG.NOM.FUT turn

mdór gónó [#37(i)]

mdó -r gónó

person gen money

“If he is buying rice, then I will be rich.”

7. [Context: same as (6), but Kucheli is buying rice.]

a. [Examples (7a-e), identical to (6a-e), were all acceptable] [#37(ii)]

To test whether the contents of the factive *sòndà* project, Yakubu was presented with (8a-e) and asked whether Kucheli and Arhel are likely to ask Musa for rice. An answer in the

⁴ Hashmarks indicate that Yakubu found the sentence unacceptable, but not ungrammatical. The absence of a checkmark indicates that Yakubu found the sentence acceptable overall.

⁵ The progressive marker *kwá* cannot co-occur with the modal *tsokta*, hence its absence from (6d).

negative would imply that the content of the complement of *sándà* (the presupposition that a dog has eaten all of the rice) has projected (see Tonhauser et al., 2013, for further examples of these indirect implication judgments, or **IJs**). Since *sándà* does not impose a **SCFC**, (8a-e) were presented in a context in which the addressees (Kucheli and Arhel) have no prior knowledge of the dog's consumption of the rice. Yakubu responded in the negative for most all variants, suggesting that the presupposition embedded under the factive *sándà* in Bura is projective, in line with data from Guaraní and English.

8. [Context: Kucheli and Arhel are looking for food to give to a family friend. Kucheli and Arhel are talking with their friend when the friend tells them about his friend Musa:]

[#37(i)]

- a. Pàzhìrna sándà àbér kila kwásmiya shìngkàfár
 pàzhì -r -na sándà àbér kila kwás -mya shìngkàfá -r

friend gen 1SG.POSS know comp dog eat compl rice gen

nì [NO]⁶ [#37(i)]

nì

3SG.POSS

“My friend knows that a dog ate all of his rice.”

⁶ [NO] indicates that Yakubu answered the IJ such that he believes Kucheli and Arhel are unlikely to ask Musa for rice. [YES/NO] indicates an ambivalent response (i.e., Kucheli and Arhel may or may not ask Musa for rice). [YES] indicates that Yakubu answered that Kucheli and Arhel would ask Musa for rice.

- b. Pàzhìrna ádí sǎndà àbér kila kwásmiya
 pàzhì -r -na ádí sǎndà àbér kila kwás -mya

friend gen 1SG.POSS not know comp dog eat compl

shìngkàfár nì wá [YES/NO] [#37(i)]

shìngkàfá -r nì wá

rice gen 3SG.POSS NEG

“My friend does not know that a dog ate all of his rice.”

- c. Pàzhìrna sǎndà àbér kila kwásmiya shìngkàfár
 pàzhì -r -na sǎndà àbér kila kwás -mya shìngkàfá -r

friend gen 1SG.POSS know comp dog eat compl rice gen

nì yá ? [NO] [#37(i)]

nì yá

3.SG.POSS q

“Does my friend know a dog ate all of his rice?”

- d. Pàzhìrna a tsokta sǎndà àbér kila kwásmiya
 pàzhì -r -na a tsokta sǎndà àbér kila kwás -mya

friend gen 1SG.POSS FUT modal know comp dog eat compl

shìngkàfár nì [YES/NO] [#37(i)]

shìngkàfá -r nì

rice gen 3.SG.POSS

“My friend could know that a dog ate all of his rice.”

associated with a belief-holder or if the content can have its effect globally (i.e., it does not have an **OLE**). As with the diagnostics for projection in **FoS**, Tonhauser et al. (2013) also provide **OLE** diagnostics curated for whether a trigger imposes a **SCFC**. These diagnostics are referenced to according to the numbering scheme in Tonhauser et al. (2013) in-line with each example. By and large, the **OLE** behavior observed for Bura tracks with equivalent data from Guaraní and English, with the sole exception of the descriptive implication of the demonstrative *ghèndá*. For instance, the expressive modifier *bzábzá* does not have an **OLE**, since the sentence in (9) is acceptable. Critically, (9) establishes that the negative evaluation of Musa is not part of the epistemic state of Arhel (who is aware of Musa’s good deeds) and is instead likely part of Kucheli’s belief-state. However, the embedding of the expressive modifier under the belief predicate *kələnkr* is still acceptable. This suggests that the expressive modifier need not have its effect locally and is apparently satisfied in the global context, since it would not be possible for Arhel to hold this descriptive content and its effective negation (i.e., the evident charity of Musa) at once in his epistemic state.

9. [Context: Kucheli, who despises her teacher Mr. Musa, is speaking with students who are going to have Mr. Musa next year but who haven’t met him yet. One of these students is Arhel. Kucheli says:]

Arhel kələnkr àbér Musa na bzábzá ná la Akindele

Arhel kələnkr àbér Musa na bzábzá ná la Akindele

name believe comp name REL expressive give DAT name

gùdùmaná kámyà ká tà nàm

gùdùmà -ná kámyà ká tà nàm

hammer 3SG.POSS because comp 3SG.NOM patch

mbwání [#41(ii)(c)]

mbwá -ní

thatch roof 3SG.ACC

“Arhel believes that damn nasty Musa gave Akindele his hammer for him to patch his thatch roof.”

The factive verb *sándà* does, by contrast, have an **OLE** (as is true for English and Guaraní). In a sentence such as (10), Kucheli believes that a dog has not eaten all of the rice (*not m*) while also believing that Arhel knows that a dog has eaten the rice (*m*). If the content of *sándà* has an **OLE**, this sentence should be judged unacceptable, given that it would require Kucheli to believe contradictory propositions. Yakubu reported that this sentence was unacceptable along these lines.

10. # Kucheli kələnkr àbór Arhel sándà àbór kila kwásmiya shìngkàfàni
 kucheli kələnkr àbór Arhel sándà àbór kila kwás -mya shìngkàfá -nì
 name believe compl name know compl dog eat compl rice def
 tsúwà Kucheli kələnkr àbór kila ádí kwásmiya shìngkàfàni wá
 tsúwà kucheli kələnkr àbór kila ádí kwás -mya shìngkàfá -nì wá
 again, also, too name believe comp dog not eat compl rice def NEG
 [#41(ii)(b)]

“Kucheli believes that Arhel knows that a dog ate all of the rice and Kucheli also believes a dog did not eat all of the rice.”

Though problematic across elicitation sessions and diagnostics, the additive particle *tsúwà* also aligned with Guaraní and English insofar as the existence implication has an **OLE**. In example (11), Musa does not know of the event of Kucheli buying rice. If *tsúwà* has an **OLE**, it would be unacceptable for Musa to believe that an ongoing alternative event is occurring, given that Musa does not know of the event in the first place. Yakubu reported that this sentence would only be acceptable if Musa knew that Kucheli was buying rice (i.e., it is unacceptable in its current formulation).

11. # Kucheli màs shìngkáfá ama Musa adí sèndà wá . Musa kələnkr àbór Arhel
 kucheli màs shìngkáfá ama Musa adí sèndà wá Musa kələnkr àbór Arhel
 name buy rice but name not know NEG name believe comp name
 màs shìngkáfá tsúwà [#41(i)]
 màs shìngkáfá tsúwà
 buy rice again, also, too
 # “Kucheli buys rice but Musa doesn’t know (this). Musa believes that Arhel buys
 rice, also.”

The only divergence from the **OLE** data reported in Guaraní and English for the Bura data was found with the descriptive implication of the demonstrative *ghèndá*, which has an **OLE**. Yakubu found (12) unacceptable and reported that it made little sense for Arhel to believe whatever was down at the river was a pigeon rather than a dog, since these two animals are markedly different. Yakubu accepted as a given that there was some referent near the river which was known to Arhel, but rejected any description by Arhel of this entity as anything other than a dog. Yakubu even questioned whether Arhel was supposed to be interpreted as being drugged or otherwise incapable of making sense, suggesting that there is a strong dispreference for a non-local satisfaction of the descriptive implication of *ghèndá*. A modification of (12), shown in (12a), was shown as an effort to test for an **OLE** of the indicational implication. Yakubu found (12a) acceptable, strengthening the possibility that the indicational implication need not have its effect locally and so does not have an **OLE** (a finding moreover in line with data from Guaraní and English).

12. # Kila kudzèhu ka zìrkù , ama Arhel adí sèndà ghènì wá . Arhel kələnkr àbér
 kila kudzèhu ka zìrkù ama Arhel adí sèndà ghènì wá Arhel kələnkr àbér
 dog near to river but name not know this NEG name believe comp
 kila ghèndá ní kàtáfár [#41(iii)]

kila ghèndá -ní kàtáfár

dog that def pigeon

“A dog is near the river but Arhel doesn’t know this. Arhel believes that that dog is a pigeon.”

a. [First sentence identical to (12), but the dog is near a house.] [#41(iii)]

8. Summary and Discussion

This investigation has tested for the acceptability of sentences containing four select triggers with particular implicational contents in Bura-Pabir, a Chadic language typologically and genetically distinct from both Guaraní and English. Each class of trigger/content proposed in Tonhauser et al. (2013) was tested for whether there a STRONG CONTEXTUAL FELICITY CONSTRAINT imposed by the trigger, whether the content was projective, and whether the content had an OBLIGATORY LOCAL EFFECT. Ultimately, the aim of this study was to produce data powering a three-way distinction between English, Guaraní, and Bura. Overall, there is an overwhelming number of correspondences between these three different languages with respect to the behavior of these four triggers – these correspondences are reflected in **Table 1**. That is, it seems that projective contents and their triggers behave similarly cross-linguistically, requiring similar context (i.e., *m*-positive utterance contexts) and having similar local effects. Though it is beyond the scope of this investigation to explore a question of semantic or broader linguistic universals, there is a suggestion in these data that languages systematically adhere to certain ‘guidelines’ with respect to projective contents. As Tonhauser et al. argue, the differences between languages may be more epiphenomenal, or at the level of inventory, rather than at nominally deeper levels of analysis (2013, p. 104).

	Properties of Contents		
	Projection	Strong Contextual Felicity	Obligatory Local Effect
Class A: additive	? YES	? YES	YES YES
Class B: expressive modifier	YES YES	NO NO	NO NO
Class B: Demonstrative NP (descriptive imp.)	YES YES	NO NO	YES NO
Class C: factive	YES YES	NO NO	YES YES
Class D: demonstrative NP (indicational imp.)	YES YES	YES YES	NO NO

Table 1. Overlap between Bura, Guaraní, and English projective contents and triggers. Red labels indicate a diagnosis from Tonhauser et al. (2013) for both Guaraní and English, while black labels refer to results from Bura. Question marks denote areas where judgments were either inconsistent or uninformative. Yellow cells indicate a contrast between Bura & Guaraní and English w/r/t projective content behavior, where green cells indicate overlap.

Naturally, this initial foray into projective contents in Bura requires further work and exploration. First and foremost, a clearer picture of the projective tendencies and contextual sensitivity of the additive *tsúwà* is needed. Some suggestions have already been offered in this regard, but a key requirement for this and other triggers is the need for more speaker data from different speakers. Tonhauser et al. (2013) relied upon the judgments of three Guaraní speakers and were largely able to minimize inconsistent judgments by incorporating multiple speakers' judgments. Following a similar logic, it may be that the inconsistencies surrounding *tsúwà* are an artifact of the interviews to be smoothed out in discussion with other speakers. Relatedly, it may be pertinent to expand the scope of this investigation to not only other implicatures covered in Tonhauser et al. (2013), but to triggers which are unique to Bura. By and large, the triggers selected for investigation here have translational equivalents in both Guaraní and English; if, in fact, the differences in projective contents are a matter of inventory, then we would still expect

similar behavior with respect to the **SCFC**, **OLE**, and projection diagnostics for items *not shared* across languages. This in turn motivates continued work with Bura, given the relative impoverishment of available lexical resources (but see Schuh & Shalanguwa, 2010). Summarily, this investigation prescribes more linguistic work on Bura-Pabir and recommends that additional speakers be sought out and presented with the data shown here. Additionally, in the name of enriching cross-linguistic comparison, more work must be done to identify potentially unique implicational triggers in Bura and subsequently test the extent to which a claim of semantic universalities in projective content behavior can be made.

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